Systems Project Management

Agile Development
Agile Project Management

Agile Software Development

- Software development methods
  - Requirements and solutions evolve
  - Collaboration between self-organizing, cross-functional teams.
  - Features
    - Adaptive planning,
    - Evolutionary development,
    - Early delivery,
    - Continuous improvement
  - Encourages rapid and flexible response to change.

Iterative, incremental and evolutionary

- Most methods break tasks into small increments with minimal planning and do not directly involve long-term planning.
- Iterations are short time frames (timeboxes) – last from one to four weeks.
  - Involves a cross-functional team working in all functions: planning, requirements analysis, design, coding, unit testing, and acceptance testing.
  - A working product is demonstrated to stakeholders.
- Minimizes overall risk and allows the project to adapt to changes quickly.
- Goal is to have an available release (with minimal bugs) at the end of each iteration.
  - An iteration might not add enough functionality to justify a market release
- Multiple iterations might be required to release a product or new features.

Efficient and face-to-face communication

- Each agile team will contain a *customer representative*, e.g. **Product Owner in Scrum**.
- This person is appointed by stakeholders to act on their behalf
  - Makes a personal commitment to being available for developers to answer mid-iteration questions.
- At the end of each iteration,
  - Stakeholders and the customer representative
    - Review progress
    - Re-evaluate priorities
- An information radiator – physical display located prominently in an office,
  - Presents an up-to-date summary of the status of a software project or other product

Very short feedback loop and adaptation cycle

- A common characteristic are daily status meetings or "stand-ups", e.g. Daily Scrum (Meeting).
- team members report to each other
  - what they did the previous day,
  - what they intend to do today
  - what their roadblocks are.

Quality focus

Specific tools and techniques are often used to improve quality and enhance project agility, such as

- Continuous integration,
- Automated unit testing,
- Pair programming,
- Test-driven development,
- Design patterns,
- Domain-driven design,
- Code refactoring

Scrum

- Developed by Ken Schwaber and Jeff Sutherland
- Name derived from Rugby
  - Group effort to move quickly to counter the opposite team, adjusting the move along progress
- Divides a project into iterations (sprints) of 30 days
  - Sprint – 30 days iteration cycle with pre-sprint and post-sprint activities
  - Define functionality for sprint
  - Leave team alone to deliver
- Sprint Goal – minimum success criterion to steer and keep focus

Scrum Lifecycle

- Planning
  - Vision, expectations, funding
- Staging
  - Identify requirements, prioritize iteration
- Development
  - Implement system ready for release in each sprint
- Release
  - Operational deployment

Scrum …

- Leaves documentation depth to specifics of projects – may need more or less
  - aim for as little as possible
- Self-directed and self-organized team
  - Competent focused people
- Demo to stakeholder at iteration end
- Client-driven adaptive planning
- No work added during iteration
- Lends itself to experimenting on certain parts of the application development

Scrum Values

- Commitment
  - Team takes responsibility to complete the Sprint. To avoid things that will stand in its way
- Focus
  - Team's focus is maintained. Distractions, interruptions are fielded
- Openness
  - Overall and individual status and commitments kept open.
- Respect
  - Team responsibility rather than accusing.
- Courage
  - Management and team have the courage to take responsibility to do what is necessary

Agile Project Steps

1. The **product owner** identifies the **product vision**.
2. The **product owner** creates a **product roadmap**.
3. The **product owner** creates a **release plan**.
4. The **product owner**, the (scrum) **master**, and the **development team** plan sprints, also called iterations, and start creating the product within those sprints
5. During each sprint, the development team has daily meetings [called scrums].
6. The team holds a sprint review.
7. The team holds a sprint retrospective.

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- The Roadmap To Value -



Release Product
[Per the Release Plan]

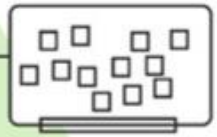
Stage 1: VISION

Description: The goals for the product and its alignment with the company's strategy.
Owner: Product Owner
Frequency: At least annually

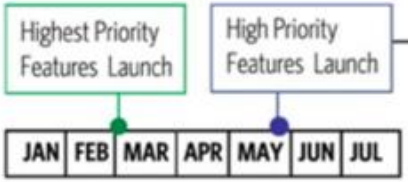


Stage 2: PRODUCT ROADMAP

Description: Holistic view of product features that create the product vision.
Owner: Product Owner
Frequency: At least biannually



Stage 3: RELEASE PLANNING



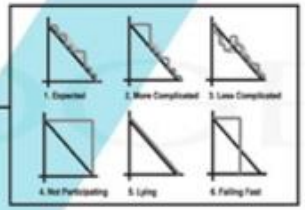
Description: Release timing for specific product functionality
Owner: Product Owner
Frequency: At least quarterly

(Stages 1-3 are best practices outside of core Scrum)



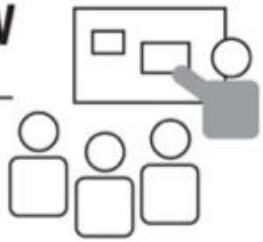
Stage 7: SPRINT RETROSPECTIVE

Description: Team refinement of environment and processes to optimize efficiency.
Owner: Scrum Team
Frequency: At the end of each sprint



Stage 6: SPRINT REVIEW

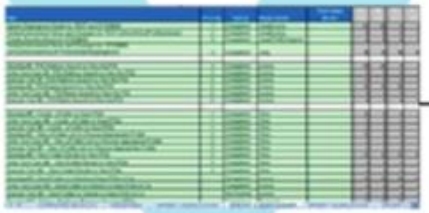
Description: Demonstration of working product
Owner: Product Owner and Development Team
Frequency: At the end of each sprint



Stage 5: DAILY SCRUM

Description: To establish and coordinate priorities of the day.
Owner: Development Team
Frequency: Daily

Stage 4: SPRINT PLANNING



Description: Establish specific iteration goals and tasks.
Owner: Product Owner and Development Team
Frequency: At the start of each sprint

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Preparation

Execution

Agile Project Artifacts

1. **Product vision statement:** An elevator pitch, or a quick summary, to communicate how your product supports the company's or organization's strategies. The vision statement must articulate the goals for the product. Revisit once a year.
2. **Product roadmap:** The product roadmap is a high-level view of the product requirements, with a loose time frame for when you will develop those requirements. Revisit twice a year.
3. **Release plan:** A high-level timetable for the release of working software.
4. **Product backlog:** The full list of what is in the scope for your project, ordered by priority. Once you have your first requirement, you have a product backlog.
5. **Sprint backlog:** The goal, user stories, and tasks associated with the current sprint.
6. **Increment:** The working product functionality at the end of each sprint.

Agile Project Roles

1. **Development team:** The group of people who do the work of creating a product. Programmers, testers, designers, writers, and anyone else who has a hands-on role in product development is a member of the development team.
2. **Product owner:** The person responsible for bridging the gap between the customer, business stakeholders, and the development team. The product owner is sometimes called a *customer representative.*
3. **Scrum master:** The person responsible for supporting the development team, clearing organizational roadblocks, and keeping the agile process consistent. A scrum master is sometimes called a project facilitator.
4. **Stakeholders:** Anyone with an interest in the project.
5. **Agile mentor:** Someone who has experience implementing agile projects and can share that experience with a project team. The agile mentor can provide valuable feedback and advice to new project teams and to project teams that want to perform at a higher level.

Agile Project Events

1. **Project planning:** The initial planning for your project.
   - includes creating a **product vision** statement and a **product roadmap,**
   - can take place in as little time as one day.
2. **Release planning:** Planning the next set of product features to release
3. **Sprint:** A short cycle of development, in which the team creates potentially shippable product functionality.
4. **Sprint planning:** A meeting at the beginning of each sprint where the scrum team commits to a sprint goal.
5. **Daily scrum:** A 15-minute meeting held each day in a sprint, where development team members state what they completed the day before, what they will complete on the current day, and whether they have any roadblocks.

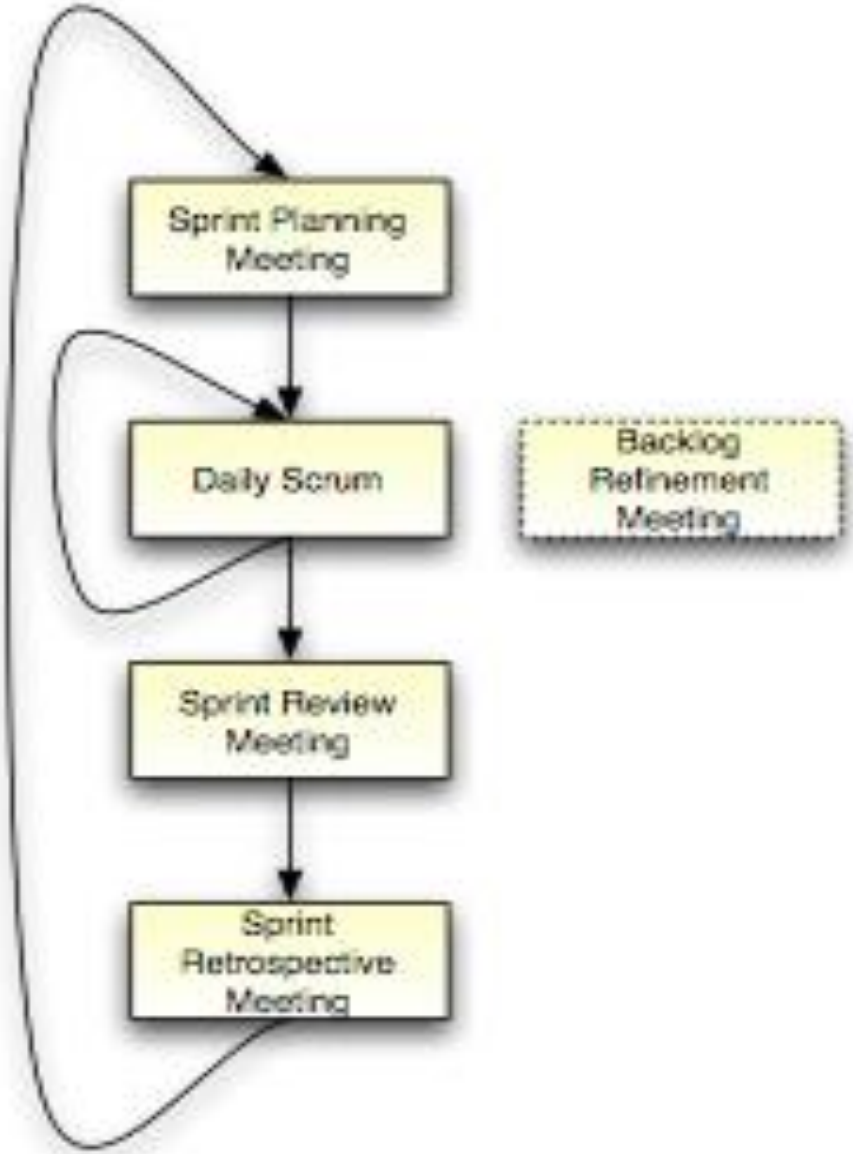
Agile Project Events

6. **Sprint review:** A meeting at the end of each sprint, where the development team demonstrates the working product functionality it completed during the sprint.
7. **Sprint retrospective:** A meeting at the end of each sprint where the scrum team discusses what went well, what could change, and how to make any changes.

Scrum

- Scrum iterations are called *sprints* and last no more than one month
- Sprints are *time-boxed*: they are never extended, but may be *restarted*
- Scrum defines three roles:
  - The *product owner* is responsible for managing the product feature list
  - The *team* does the work
  - The *scrum master* facilitates the scrum process; the scrum master is *not* a ‘conventional’ project manager
- Scrum produces an executable, *potentially shippable product* increment at the end of each sprint
  - *Potentially shippable* means ‘done’ in the scrum sense: all commitments for the sprint have been met – no “it’s 80% done” is allowed
  - This demonstrates value iteratively to the product owner and to other stakeholders

Scrum Meetings



- Planning meeting
- Sprint
  - Scrum meeting
- Sprint review
- Sprint retrospective

The Sprint

- **Planning meeting** at the start of the sprint
  - create a sprint backlog – a list of the tasks to perform during the sprint.
- During the sprint, the team takes a small set of features from idea to coded and tested functionality.
- At the end, these features are done, meaning coded, tested and integrated into the evolving product or system.

Backlog

- Backlog – used for planning
- Complete functionality that remains to be added to the product.
  - Features and estimate of duration for each task
  - Tasks for sprint picked from the pool of tasks
  - Used to decide features for sprint and plan out the work

Scrum

- Every day the team holds a short (fifteen minute) meeting, called a scrum,
  - Scrum meeting – Short standup meeting to communicate and monitor progress
  - Daily scrums as a way to synchronize the work of team
  - share what they worked on the prior day,
  - The team runs through what it will do in the next day
  - Surfaces roadblocks
  - Some recommend all participants stand
- Scrum Master – coach for the team
  - Looks outward keeping distractions out
  - Trusts the self-managed team to get work done

The Sprint

The Sprint Review

- The team demonstrates the new functionality to the PO or any other stakeholder who wishes to provide feedback that could influence the next sprint.
- This feedback loop within Scrum software development may result in changes to the freshly delivered functionality, but it may just as likely result in revising or adding items to the product backlog.

The Sprint

- The **sprint retrospective** is at the end of each sprint. The whole team participates in this meeting, including the Scrum Master and PO.
- The meeting is an opportunity to reflect on the sprint that has ended, and identify opportunities to improve.
- The team reflects on its own process. They inspect their behavior and take action to adapt it for future Sprints.
- A common impediment to full transparency on the team is the presence of people who conduct performance appraisals.
- Retrospectives often expose organizational impediments.

Agile Project Management

How are Agile Projects Managed?

Scrum

- On a Scrum project, there are three roles:
  - Product Owner
  - Team
  - Scrum Master

The Product Owner

- Responsible for the business aspects of the project, including ensuring the right product is being built, and in the right order.
  - Balance competing priorities,
  - Available to the team,
  - Empowered to make decisions about the product.
- Single person responsible for maximizing the return on investment (ROI) of the development effort
- Responsible for product vision
- Constantly re-prioritizes the Product Backlog, adjusting any long-term expectations such as release plans
- Final arbiter of requirements questions

The Product Owner (cont.)

- Accepts or rejects each product increment
- Decides whether to ship
- Decides whether to continue development
- Considers stakeholder interests
- May contribute as a team member
- Has a leadership role

The Team

- Cross-functional (e.g., includes members with testing skills, and often others not traditionally called developers: business analysts, domain experts, etc.)
- Self-organizing / self-managing, without externally assigned roles
- Negotiates commitments with the Product Owner, one Sprint at a time
- Has autonomy regarding how to reach commitments
- Intensely collaborative

The Team (cont.)

- Most successful when located in one team room, particularly for the first few Sprints
- Most successful with long-term, full-time membership. Scrum moves work to a flexible learning team and avoids moving people or splitting them between teams.
- 7 ± 2 members
- Has a leadership role

The Scrum Master

- Facilitates the Scrum process
- Helps resolve barriers/blocks
- Creates an environment conducive to team self-organization
- Captures empirical data to adjust forecasts
- Shields the team from external interference and distractions to keep it in group flow (a.k.a. the zone)
- Enforces timeboxes
- Keeps Scrum artifacts visible
- Promotes improved engineering practices
- Has no management authority over the team (anyone with authority over the team is by definition not its Scrum Master)
- Has a leadership role

The Scrum Master

- Unlike a traditional project manager, the Scrum Master is not viewed as the person to credit (or blame) for the success (or failure) of the project.
- The Scrum Master's authority extends only to the process. The Scrum Master is an expert on the process, and on using it to get a team to perform to its highest level.
- A Scrum Master does not have many of the traditional responsibilities – scope, cost, personnel, risk management – that a project manager does.

The Best Agile Methodologies



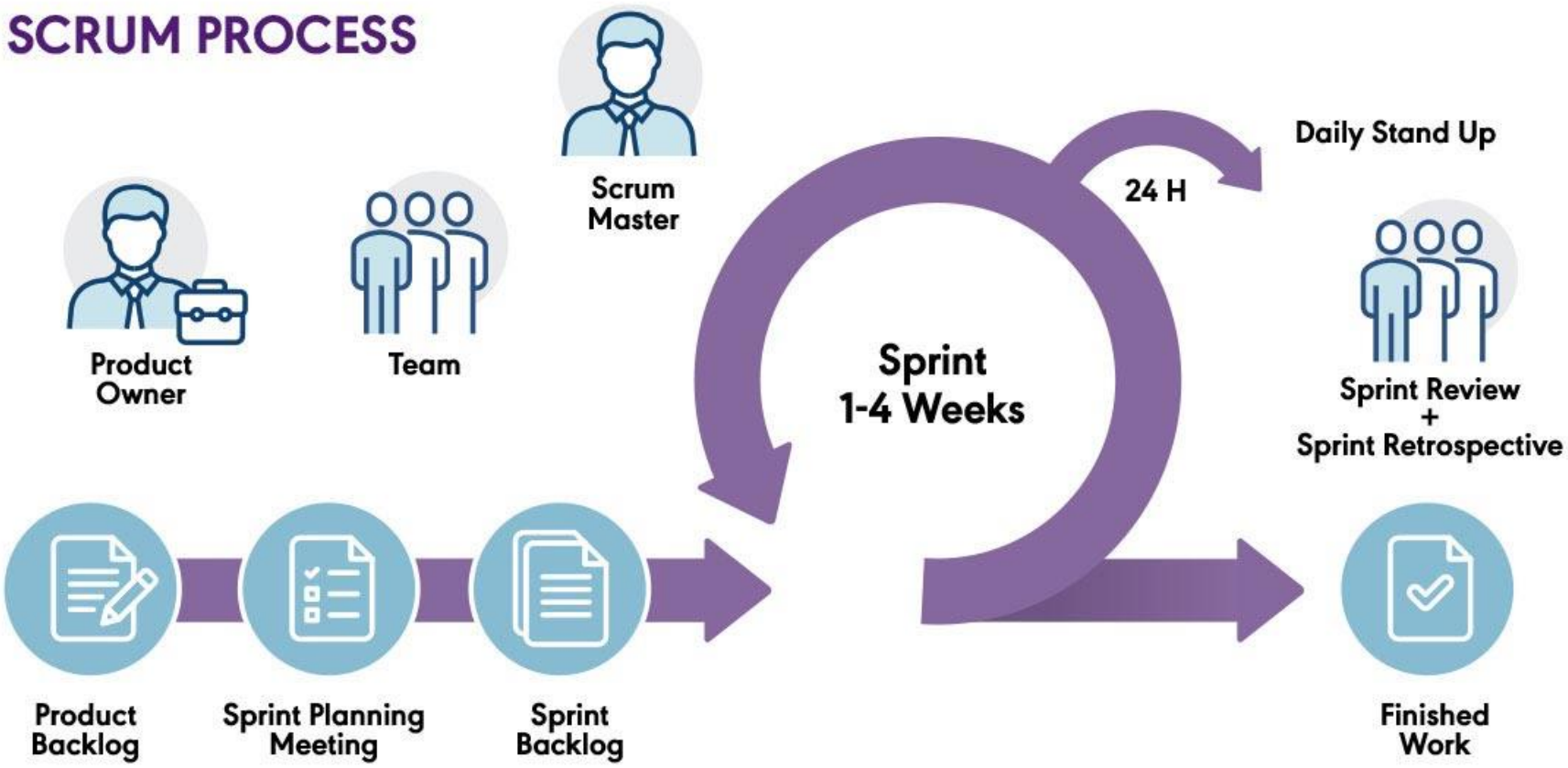
Agile Methodologies

1. Scrum
2. Kanban
3. Scrumban
4. Extreme Programming (XP)
5. Lean Software Development
6. Crystal methodology
7. Dynamic Systems Development Method (DSDM)
8. Feature Driven Development (FDD)
9. Adaptive Software Development (ASD)

Scrum Method

- Scrum is the most popular Agile framework, defined by the Scrum Guide in 2010. According to the 15th State of Agile report, 66% of respondents identified Scrum as one of the Agile frameworks they follow most closely.
- **What are the main characteristics of a Scrum team?**
  - Scrum teams work in short iterations called Sprints, with 2 weeks being the most popular choice.
  - Each Sprint has its goal.
  - Scrum clearly defines the team roles like Scrum Master, Product Owner, and Development Team.
  - Scrum teams attend regular Agile events, like Sprint Planning, Daily Scrum, Sprint Review, and Sprint Retrospective.

Scrum Method



Which teams would benefit the most from Scrum?

- Scrum works well in companies with **frequent cross-team collaboration**.
- It fits **small teams, 3 to 9 people maximum**.
- It's a good choice for teams that work in **short development cycles** focused on **timely releases**.
- The Scrum framework is also great **for "newbies" in Agile** because it provides **clear guidelines**.
- It's one of the best-defined Agile frameworks: **a good choice to start your Agile journey.**

Kanban Method

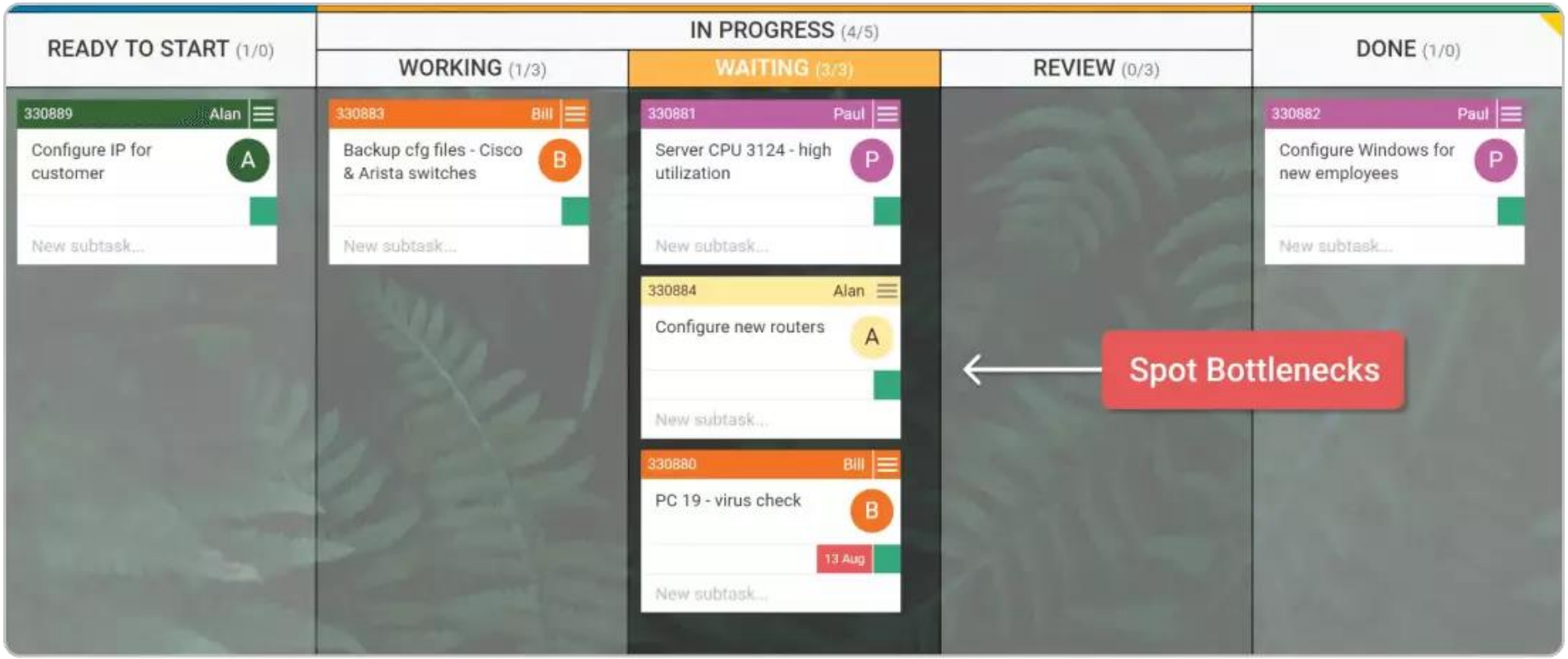
- Kanban is roughly translated from Japanese and means “cards you can seeIn Kanban, the progress of the development process is visualized on a Kanban board.

- **What are the main characteristics of a Kanban team?**

- Kanban teams work with a Kanban board. Typically the Kanban board includes the “To Do,” “In Progress,” and “Done” columns.
- In software development, the Kanban board is usually extended to: “To Do,” “Dev,” “Test,” and “Done.”
- Kanban teams work with **“Work in Progress” limits to prevent bottlenecks** and achieve quicker release cycles.
- Kanban is a **continuous workflow**. Product improvements happen on an ongoing basis through **continuous collaboration** and **ad-hoc team meetings**.

Kanban Board

1. The best way to visualize our work is by creating and using a Kanban board.
2. The simplest one may consist of three columns – “Requested”, “In Progress,” and “Done”.



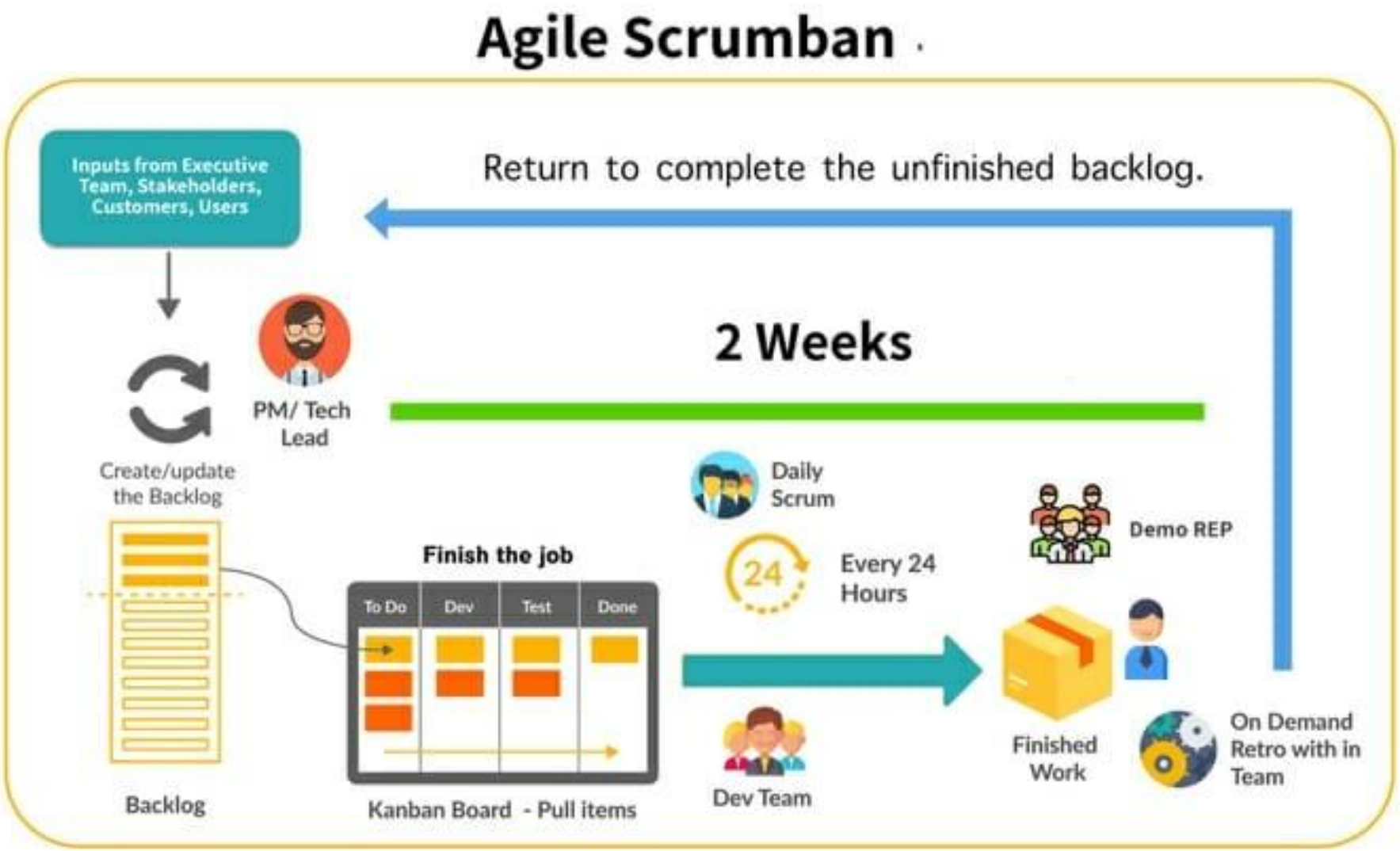
Which teams would benefit the most from Kanban?

- Kanban is the way to go for teams that handle **different-size tasks** and **need to juggle frequently changing product** requirements.
- Kanban methodology is not as clearly defined as Scrum, so it **wouldn't be advised to inexperienced Agile teams**.
- The Kanban concept is usually recommended to smaller teams that work on **repetitive tasks** and are **not heavily dependent on others**.
- Kanban is a great choice **if your team prioritizes the speed to market and is more advanced in Agile methods**.

Scrumban Method

- Scrumban is a Scrum Kanban hybrid, taking some of the principles from Scrum and some of the principles from Kanban.
- **Main characteristics of the Scrumban team:**
- Scrumban teams work on a time-based schedule. Instead of 2-week Sprints, they work with 3-month, 6-month, and 1-year buckets.
- Agile events are not a must in Scrumban. The team can choose to skip certain events, however, retro sessions and ad-hoc planning are definitely recommended.
- Scrumban teams visualize their tasks on a Scrumban board with applied 'Work In Progress' limits.

Scrumban Method



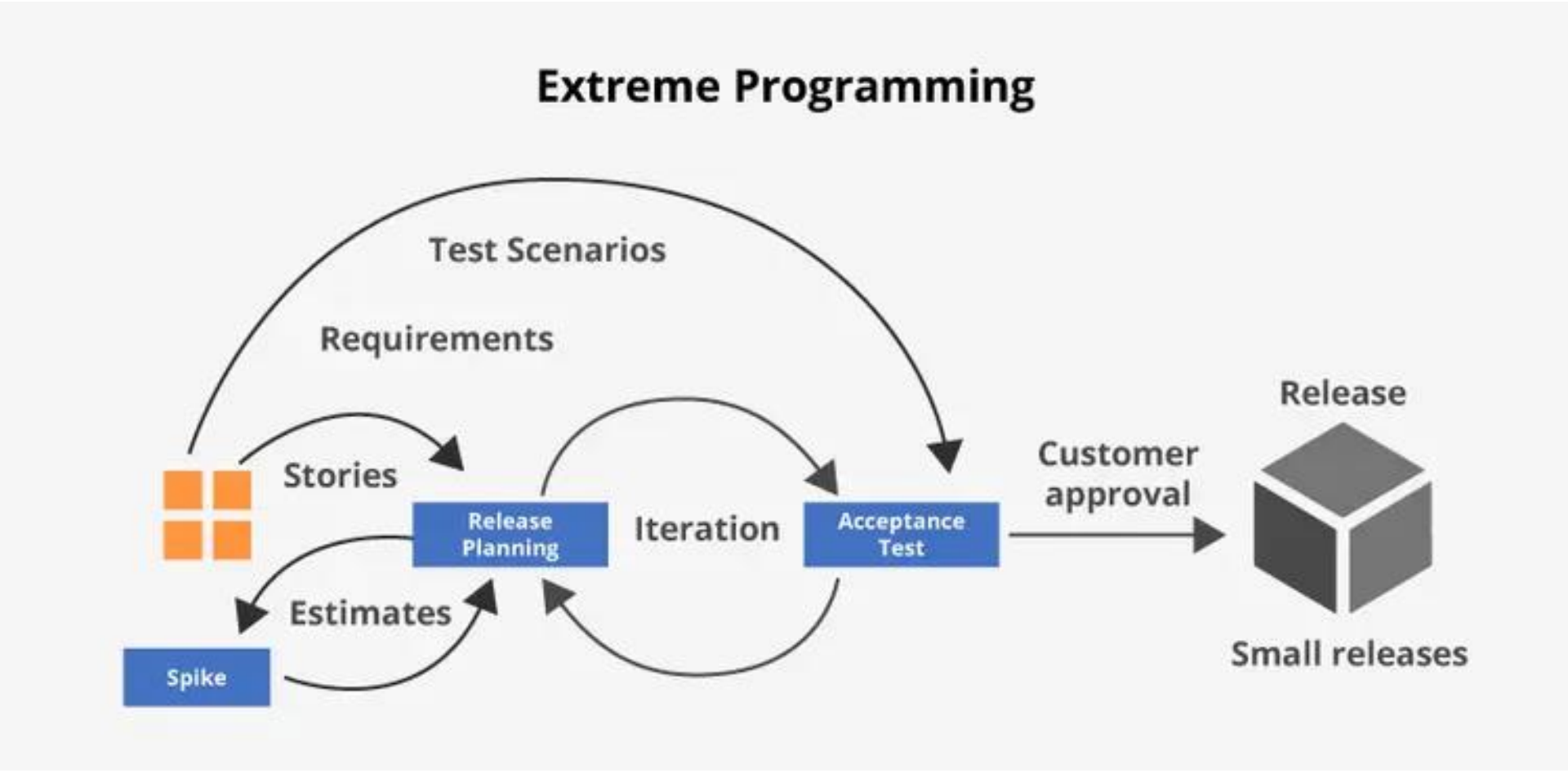
Which teams would benefit the most from Scrumban?

- Scrumban is a great choice for either small or large teams that are experienced or just starting with Agile.
- It's a good fit for those who find Scrum too rigid and think that Scrum rules are hindering the team's productivity.
- Scrumban works well with teams that are developing multiple products in fast-paced environments and have less strict estimation requirements from the senior management.

Extreme Programming (XP) Method

- Extreme Programming (XP) focuses on producing higher quality software, sharing knowledge, and taking care of the software teams' wellbeing.
- **main characteristics of the EP team?**
- The team values simplicity, quick feedback, collaboration, and quality work.
- XP teams go through 5 development phases: planning, designing, coding, testing, and listening.
- The most important practices are test-driven development, pair programming, simple design, collective code ownership, plus a 40-hour week that prevents burnout.

Extreme Programming (XP) Method



Which teams would benefit the most from XP?

- XP is a good choice for mixed-skill level teams, especially ones where junior and senior programmers work together.
- It could also work well for teams with tight deadlines, smaller budgets, and frequently changing project requirements.
- Extreme programming would not be recommended to remote teams, as it performs best when applied to small teams that work from the same location.

Lean Software Development

- Lean product development focuses on producing only what the product actually needs. It optimizes time and resources and reduces unnecessary activities.

- **Main characteristics of the Lean Software Development team?**

- The team works with Minimum Viable Products (MVPs) that are released to the customers as early as possible. Customer feedback is then collected and included in future software iterations.
- The team eliminates redundant activities to save time (like reworking the code or using resources in an inefficient way), and team members are empowered to make their own decisions, which results in higher morale.

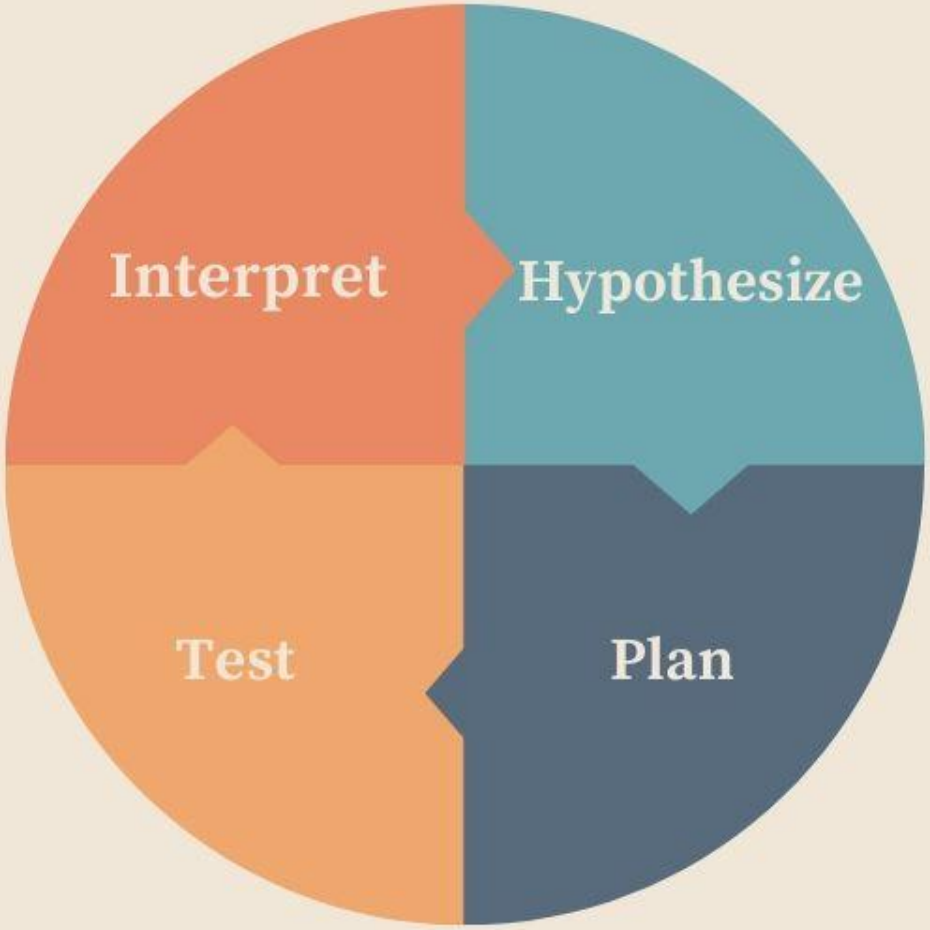
The 4 Stages of Lean Product Development

Evaluate data to analyze and synthesize results



Draw conclusions and make predictions





Run experiments and gather data



Incorporate design changes



Which teams would benefit the most from Lean?

- Teams that are responsible, experienced, and can be trusted to make important decisions on their own would benefit the most from the Lean methodology.
- Lean methodology is not easily scalable as it's more suited to smaller teams that are cohesive(the action or fact of forming a united whole) as well as teams that can embrace(hold) more detailed documentation.

Crystal methodology

- Crystal methodology is made up of smaller Agile project management methodologies that were first introduced by Dr. Alistair Cockburn, co-author of the Agile Manifesto, and place the main focus on collaboration practices.

- **main characteristics of the teams for Crystal approach?**

- Teams following the Crystal approach work with smaller Agile development methodologies like Crystal Clear, Crystal Yellow, Crystal Orange, Crystal Orange Web, Crystal Red, Crystal Maroon, Crystal Diamond, and Crystal Sapphire.
- The Crystal approach is very flexible and adaptable, and it works well for different types of Agile teams.
- Crystal teams avoid unnecessary documentation and reporting and practice one of the lightweight Agile software development methodologies.

Crystal methodology

Method	Team size	Project size
Crystal Clear	6 people or less	Small projects
Crystal Yellow	7–20 people	Small to medium projects
Crystal Orange	20–40 people	Medium projects
Crystal Red	40–80 people	Medium to large projects
Crystal Maroon	80–200 people	Large projects
Crystal Diamond **or** Crystal Sapphire	Very large teams (200+ people)	Very large projects with high criticality

Which teams would benefit the most from Crystal?

- Different team sizes can benefit from the Crystal project management process. For example, Crystal Clear fits smaller teams and shorter projects (up to 6 developers), and Crystal Red works better with larger teams (up to 80 team members).
- The Crystal approach is a good choice for teams that communicate frequently and cooperate without the day-to-day involvement from the higher management.
- Crystal methodologies are not recommended to inexperienced Agile teams as the structure is not clearly defined, and there are some grey areas open to interpretations.

Dynamic Systems Development Method (DSDM)

- Created in 1994, DSDM was meant to complement another Agile framework RAD (Rapid Application Development). DSDM's main focus was to help with planning and creating more discipline in the team.

- **What are the main characteristics of the team following the Dynamic Systems Development Framework?**

- Timely delivery, focus on business needs, and not compromising on code quality.
- DSDM teams work on a fixed cost, fixed time schedule, and negotiable features.
- Teams following the DSD framework work mostly with the **MoSCoW** prioritization technique, facilitated workshops, and time-boxed iterative development.

Which teams would benefit the most from DSDM?

- DSDM Agile framework works well with small as well as larger, more complex projects.
- It would be a good fit for teams that can embrace ambiguity and frequently changing requirements.
- DSDM works well in the corporate environment, where other Agile methodologies can be harder to implement.

Feature Driven Development (FDD)

- Feature Driven Development originated from a long 15 - month project, in a Singaporean bank, back in 1997. FDD blends different types of Agile methodologies and puts customer satisfaction at the center.

- **What are the main characteristics of the team following the Feature Driven Development method?**

- Teams adopting Feature Driven Development focus mainly on making progress on features.
- FDD teams follow a 5-step development process flow: “develop an overall model,” “build a feature list,” “plan by feature,” “design by feature,” and “build by feature.”
- Features that the FDD teams build are similar to the user stories we know from Scrum.

Which teams would benefit the most from FDD?

- FDD Agile method is handy in large-scale software projects, especially in the finance and banking industry, where the main focus is on speeding up the release of features.
- It wouldn’t be recommended for smaller projects, though.
- You might want to test it out if your project grows too complex for the regular Scrum team to handle, especially in organizations where only software development teams follow Agile methodologies.

Adaptive Software Development (ASD)

- Adaptive Software Development has also originated from RAD. The main focus of ASD is to adapt rapidly to changing product requirements or sudden shifts in user behavior and market trends.

- **Main characteristics of the team for ASD methodology?**

- The team develops software according to three phases: speculate, collaborate, and learn.
- The ASD team focuses mainly on the end user and user feedback which can lead to more user-friendly products.
- Teams are required to deliver on time, embracing early delivery.
- Testing is introduced into every stage of the development process.

Which teams would benefit the most from ASD?

- If your company puts the main emphasis on timely delivery and is expected to continuously change the offering, you might want to test the Adaptive Software Development methodology with your team.
- The ASD method works well for teams involved in complex software projects; however, it might not be the best choice if there is no room or budget to introduce user testing into every stage of the development process.

Agile Project Management Cheat Sheet:

Which Method is Right for You?

Agile Project Management Cheat Sheet

Agile Methodology	Suitable For	Known For
Scrum	· Small teams · Cross-team collaboration	· Scrum roles · Sprints · Agile events
Kanban	· Advanced Agile teams · Frequently changing product requirements	· Kanban board · Continuous workflow
Scrumban	· Fast-paced environment · Developing multiple products	· Scrumban board · 3-month, 6-month & 1-year buckets
Extreme Programming	· Same location · Mixed-skill level teams · Smaller budgets	· Test-driven development · Pair programming · Simple design · Collective code ownership
Lean	· Small, empowered teams that make their own decisions	· MVP · Eliminating redundant activities

Agile Project Management Cheat Sheet

Agile Methodology	Suitable For	Known For
Crystal	• Different team sizes • Advanced teams in Agile • Less involvement from the higher management	• Crystal Clear, Crystal Yellow, Crystal Orange, Crystal Orange Web, Crystal Red, Crystal Maroon, Crystal Diamond, and Crystal Sapphire.
DSDM	• Frequently changing requirements • Corporate environment	• Fixed cost, fixed time schedule, negotiable features • MoSCoW prioritization technique
FDD	• Large-scale software projects • Finance and banking industry	• Making progress on features • 5-step development process flow
ASD	• Timely delivery • Continuously updated offering • Complex software projects	• Three phases: speculate, collaborate, and learn • Early delivery • Frequent testing

Agile Methodologies Used

Scrum and related variants continue to be the most common Agile methodologies used by respondents' organizations.

